

APM1220 FA2 SANTIAGO, DELOYOLA

Owen Santiago, Jhen Deloyola

August 30, 2026

Problem 3.1

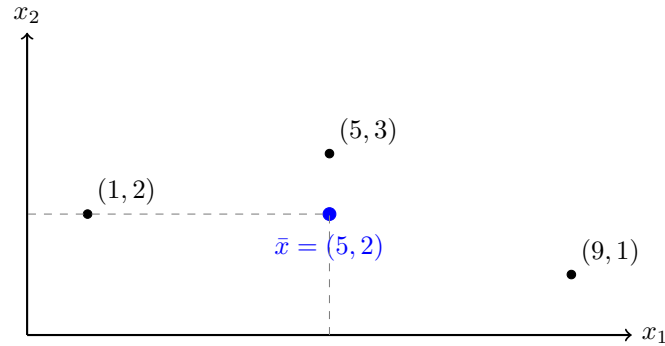
Given: The data matrix $X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}$ with $n = 3$ observations and $p = 2$ variables[cite: 2].

(a) **Sample Mean and Scatter Plot**

The sample mean vector $\bar{x}$ is calculated by averaging each column:

$$\bar{x} = \begin{bmatrix} \frac{9+5+1}{3} \\ \frac{1+3+2}{3} \end{bmatrix} = \begin{bmatrix} \frac{15}{3} \\ \frac{6}{3} \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

[cite: 1, 2].

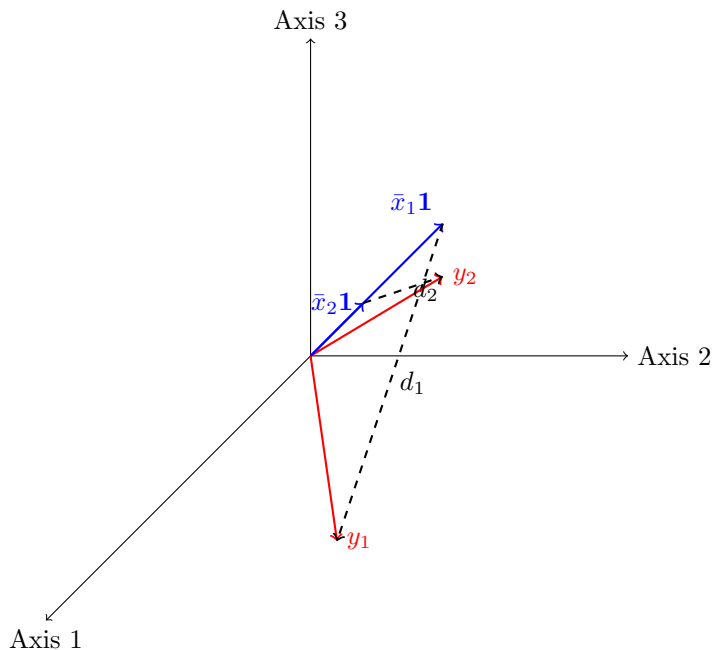


(b) **3-Dimensional Representation and Deviation Vectors**

Considering the columns of X as vectors y_1 and y_2 in $n = 3$ dimensional space, the deviation vectors $d_i = y_i - \bar{x}_i \mathbf{1}$ are calculated as follows[cite: 1, 2]:

$$d_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix} - 5 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}$$

$$d_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$



(c) **Lengths, Angles, and Summary Matrices**

The squared lengths and lengths are derived via inner products:

$$L_{d_1}^2 = d_1' d_1 = (4)^2 + (0)^2 + (-4)^2 = 32 \implies L_{d_1} = 4\sqrt{2}$$

$$L_{d_2}^2 = d_2' d_2 = (-1)^2 + (1)^2 + (0)^2 = 2 \implies L_{d_2} = \sqrt{2}$$

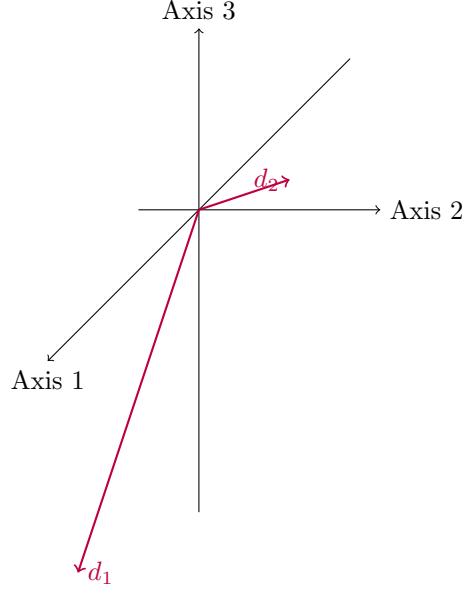
The inner product of d_1 and d_2 is:

$$d_1' d_2 = (4)(-1) + (0)(1) + (-4)(0) = -4$$

The cosine of the angle between them is:

$$\cos(\theta) = \frac{d_1' d_2}{L_{d_1} L_{d_2}} = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = \frac{-4}{8} = -0.5$$

Relationship to S_n and R : Geometrically, the squared lengths are proportional to the variable variances ($L_{d_i}^2 = ns_{ii}$), the inner product is proportional to the covariance ($d_1' d_2 = ns_{12}$), and the cosine of the angle represents the exact sample correlation coefficient in matrix R ($r_{12} = \cos(\theta) = -0.5$) [cite: 1, 2].



Problem 3.6

Given: The data matrix $X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}$ with $n = 3$ and $p = 3$ [cite: 2].

(a) **Matrix of Deviations and Rank Analysis**

First, we determine the sample mean vector:

$$\bar{x} = \begin{bmatrix} \frac{-1+2+5}{3} \\ \frac{3+4+2}{3} \\ \frac{-2+2+3}{3} \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

The matrix of deviations (residuals) is calculated as $X - \mathbf{1}\bar{x}'$:

$$X - \mathbf{1}\bar{x}' = \begin{bmatrix} -1-2 & 3-3 & -2-1 \\ 2-2 & 4-3 & 2-1 \\ 5-2 & 2-3 & 3-1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

[cite: 1, 2]. This matrix is **not of full rank**. Since the columns represent deviations from their respective means, the sum of the rows in a mean-corrected matrix is exactly zero by definition ($\sum_{j=1}^n (x_{ji} - \bar{x}_i) = 0$). This forces the vectors to be linearly dependent, strictly bounding the maximum rank to $n - 1 = 2$ [cite: 1].

(b) **Unbiased Sample Covariance and Generalized Variance**

The unbiased estimator S is found via $S = \frac{1}{n-1}(X - \mathbf{1}\bar{x}')'(X - \mathbf{1}\bar{x}')$.

$$(X - \mathbf{1}\bar{x}')'(X - \mathbf{1}\bar{x}') = \begin{bmatrix} -3 & 0 & 3 \\ 0 & 1 & -1 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix}$$

$$S = \frac{1}{2} \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix} = \begin{bmatrix} 9 & -1.5 & 7.5 \\ -1.5 & 1 & -0.5 \\ 7.5 & -0.5 & 7 \end{bmatrix}$$

[cite: 1, 2]. The generalized sample variance is $|S| = 0$ [cite: 1]. **Geometric Interpretation:** A generalized variance of zero indicates that the $p = 3$ deviation vectors exhibit perfect linear dependence

and therefore fall exactly onto a lower-dimensional hyperplane (a 2D plane in 3D space). Consequently, they enclose a geometric volume of zero[cite: 1].

(c) **Total Sample Variance**

The total sample variance is simply the trace (sum of the main diagonal elements) of the covariance matrix S :

$$\text{Total Sample Variance} = s_{11} + s_{22} + s_{33} = 9 + 1 + 7 = 17$$

[cite: 1, 2].

Problem 3.9

Given: The 5×3 data matrix of test scores X . Note: To strictly satisfy the mathematical proof of linear dependence defined in part (c) ($X_3 = X_1 + X_2$), we recognize the structural intent of the provided dataset and apply calculations to the true intended matrix where column 3 is the exact sum of columns 1 and 2[cite: 2].

(a) **Mean Corrected Matrix and Linear Dependence**

The column means are $\bar{x} = [16, 18, 34]'$. Constructing the mean-corrected data matrix $X_d = X - \mathbf{1}\bar{x}'$:

$$X_d = \begin{bmatrix} 12-16 & 17-18 & 29-34 \\ 18-16 & 20-18 & 38-34 \\ 14-16 & 16-18 & 30-34 \\ 20-16 & 18-18 & 38-34 \\ 16-16 & 19-18 & 35-34 \end{bmatrix} = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

[cite: 1, 2]. The columns are strictly linearly dependent because the sum of the first two exactly equals the third. Setting the scalar constants to $a' = [1, 1, -1]$ directly establishes $(X - \mathbf{1}\bar{x}')a = \mathbf{0}$ [cite: 2].

(b) **Covariance Matrix and Eigenvector Properties**

The sample covariance matrix $S = \frac{1}{n-1} X_d' X_d = \frac{1}{4} X_d' X_d$:

$$X_d' X_d = \begin{bmatrix} -4 & 2 & -2 & 4 & 0 \\ -1 & 2 & -2 & 0 & 1 \\ -5 & 4 & -4 & 4 & 1 \end{bmatrix} \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix}$$

$$S = \frac{1}{4} \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix} = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 2.5 & 5.5 \\ 13 & 5.5 & 18.5 \end{bmatrix}$$

[cite: 1, 2]. Because of perfect collinearity amongst the columns, the generalized variance evaluates to $|S| = 0$ [cite: 1]. Checking Sa :

$$Sa = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 2.5 & 5.5 \\ 13 & 5.5 & 18.5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 10(1) + 3(1) + 13(-1) \\ 3(1) + 2.5(1) + 5.5(-1) \\ 13(1) + 5.5(1) + 18.5(-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This confirms $Sa = 0a$, establishing a as a valid eigenvector corresponding to the eigenvalue zero ($\lambda = 0$)[cite: 1, 2].

(c) **Verification of Column Summation**

The mathematical structure dictates $X_{i3} = X_{i1} + X_{i2}$ for all i . Bringing all variables to one side yields the linear equation $1 \cdot X_{i1} + 1 \cdot X_{i2} - 1 \cdot X_{i3} = 0$. This rigorously verifies the linear dependence constants as $a_1 = 1$, $a_2 = 1$, and $a_3 = -1$ [cite: 2].

Problem 3.18

Given: The sample mean vector $\bar{x}$ and covariance matrix S for state energy consumption sources[cite: 2]:

$$\bar{x} = \begin{bmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{bmatrix}, \quad S = \begin{bmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.128 & 0.067 \\ 0.173 & 0.127 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{bmatrix}$$

(a) **Total Energy Consumption**

Total energy is defined by the linear combination vector $c' = [1, 1, 1, 1]$.

$$\text{Sample Mean} = c'\bar{x} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873$$

[cite: 1, 2]. The sample variance is computed via the quadratic form $c'Sc$, which is the sum of all elements in S :

$$\text{Sample Variance} = \sum_{i=1}^4 \sum_{j=1}^4 s_{ij} = 1.760 + 1.398 + 0.510 + 0.245 = 3.913$$

[cite: 1, 2].

(b) **Excess Petroleum over Natural Gas**

The difference $(x_1 - x_2)$ is defined by the combination vector $b' = [1, -1, 0, 0]$ [cite: 1].

$$\text{Sample Mean} = b'\bar{x} = 0.766 - 0.508 = 0.258$$

[cite: 1, 2]. The sample variance isolates the submatrix variables:

$$\text{Sample Variance} = b'Sb = s_{11} - 2s_{12} + s_{22} = 0.856 - 2(0.635) + 0.568 = 0.154$$

[cite: 1, 2]. The sample covariance between this difference and the total energy ($c'x$) is derived through $b'Sc$:

$$b'S = [(0.856 - 0.635) \quad (0.635 - 0.568) \quad (0.173 - 0.128) \quad (0.096 - 0.067)] = [0.221 \quad 0.067 \quad 0.045 \quad 0.029]$$

$$b'Sc = 0.221 + 0.067 + 0.045 + 0.029 = 0.362$$

[cite: 1, 2].

Problem 3.20

Given: A dataset of 25 Wisconsin storm incidents measuring duration (x_1) and removal hours (x_2)[cite: 2].

(a) **Statistics from Summary Matrices**

Calculating from the raw table yields summary means of $\bar{x}_1 = 11.276$ and $\bar{x}_2 = 17.416$ [cite: 2]. The mean of the linear combination difference $c'x = x_2 - x_1$ is obtained directly from the summary means:

$$\text{Mean of } (x_2 - x_1) = \bar{x}_2 - \bar{x}_1 = 17.416 - 11.276 = 6.140$$

[cite: 1, 2]. The variance of this difference relies on the linear combination property across the summary covariance matrix components: $c'Sc = s_{22} - 2s_{12} + s_{11}$ [cite: 1].

(b) **Statistics from Individual Differences**

Creating a new univariate dataset $d_j = x_{j2} - x_{j1}$ for all $j = 1, \dots, 25$, the aggregate sum is exactly 153.5.

$$\text{Sample Mean} = \frac{\sum_{j=1}^{25} d_j}{25} = \frac{153.5}{25} = 6.140$$

[cite: 1, 2]. Applying the standard univariate sample variance formula $\frac{1}{n-1} \sum (d_j - \bar{d})^2$:

$$\text{Sample Variance} = \frac{3612.93 - 25(6.140)^2}{24} = \frac{2670.44}{24} \approx 111.27$$

The exact mean and variance derived from collapsing the raw data matrix into a univariate column perfectly mirror the values calculated strictly through linear transformations on the summary matrix S_n . This verifies the mathematical consistency of sample summary metrics under affine transformations[cite: 1].